Sahaj Bhadja

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Summary

Software engineer with 2+ years building reliable technical systems, simulation pipelines, automated tests, and data-processing applications. Skilled in Python, C++, Java, Linux, debugging, CI/CD, visualization, and cross-functional engineering delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected multi-tenant engineering services on Azure SQL and Blob Storage for 500+ users, applying normalized models, validation, monitoring, and indexed queries to reduce analytical dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant asynchronous workflows for 1,000+ monthly jobs with explicit state transitions, retries, failure isolation, and operational monitoring, improving throughput by 3x while preventing cascading processing failures.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered Python services and data-ingestion interfaces handling 10,000+ daily requests, integrating secure uploads, AWS RDS persistence, and modular processing while maintaining stable production performance during peak usage.
- Implemented concurrent processing for 1,000+ files with validation, error isolation, structured logging, code reviews, and automated tests, reducing experimental processing turnaround by 60% for 150+ users.

USC AD Labs **Los Angeles, CA**
Machine Learning Researcher *January 2025 – May 2025*
- Built a Python simulation pipeline processing 100,000+ trajectory logs, improving long-horizon autonomous-driving planning by 18% through offline reinforcement learning, reproducible evaluation, data visualization, and structured experiment tracking.
- Architected a spatial-temporal encoder combining HD-map geometry and agent trajectories through multi-head attention, improving closed-loop simulation stability by 15% while reducing off-road behavior during comprehensive system-level testing.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Engineered cross-platform software with object-oriented design, Jenkins pipelines, automated VM builds, and 20+ PyTest and Mockito suites across Windows, Linux, and macOS, strengthening verification and production reliability.
- Optimized API and data-processing workflows using AWS Glue, Amazon S3, SQL, indexing, caching, and pagination, improving application response times by 30% and production pipeline efficiency by 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed real-time application workflows with REST APIs and WebSockets, synchronizing 2M+ records across modular services while reducing conflicts through validation controls, fault handling, and deterministic state updates.
- Automated Docker-based builds and deployment pipelines using Jenkins CI/CD, unit validation, and release checks, reducing deployment time by 50% while improving repeatability across development and production environments.

Skills

Languages & Systems: Python, C++, Java, JavaScript, SQL, Bash, Linux, UNIX, object-oriented programming
Data & Automation: data acquisition, data processing, ETL pipelines, visualization, background workers, REST APIs, WebSockets
Testing & Reliability: unit testing, integration testing, PyTest, Mockito, debugging, root-cause analysis, monitoring, validation
Tools & Infrastructure: Git, GitHub, Jenkins, CI/CD, Docker, Kubernetes, IDEs, compilers, AWS, Azure, PostgreSQL
Engineering Practices: algorithms, data structures, distributed systems, system design, technical documentation, code reviews, Agile, deployment, operational support

Projects

CPU Scheduling Simulator | *C++, React, JavaScript, EJ-2*
- Implemented six scheduling algorithms with deterministic simulation, interactive parameter controls, edge-case testing, and Gantt visualization across 1,000+ process combinations, reducing debugging time by 20% while validating timing metrics.

Linguistix | *Python, Linux, AWS, Docker, CodePipeline*
- Developed a containerized data-processing application with automated releases, validation checks, structured error handling, and scalable interfaces, achieving 87% accuracy while strengthening reproducible deployment workflows and production reliability.

CaptionCraft | *Python, PyTorch, Transformers, Flask*
- Benchmarked BLIP, GIT, and ViT-GPT-2 architectures across 10,000 images and 100,000 captions, improving accuracy from 81% to 96% while building tested inference interfaces for real-time technical evaluation.